

Quantifying Model, Label, and Analyst Contributions to Variance in Real-Time Sepsis Onset Prediction

Abdul Ahad

M.Sc. in Computing

Department of Computer Science & Applied Physics

Atlantic Technological University (ATU), Galway

I. Introduction: The Clinical Problem

- **Sepsis** is a life-threatening condition caused by the body's extreme response to an infection [1].
- It causes **about 11 million deaths every year**, nearly **1 in 5 deaths worldwide** [2].
- Delaying treatment by even a few hours increases the risk of death [3].
- Early warning systems aim to detect sepsis before doctors recognise it, allowing earlier treatment.

More than 90 AI models have already been developed to predict sepsis [4].

Existing models have improved prediction, but important methodological questions remain unanswered.

I. Introduction: Research Question

RQ

In real-time sepsis onset prediction on MIMIC-IV, how much of the observed performance variation is explained by the choice of model, the sepsis label definition, and the analyst's methodological decisions?

II. Background: The Gap Between Predictive Performance and Clinical Utility

Wang et al. [4] reviewed 91 real-time sepsis prediction models and found:

Metric	Internal	External
Median AUROC	0.811	0.783
Median Utility Score	+0.381	-0.164

- AUROC stays almost the same, so the model looks good.
- But the Utility Score becomes negative, meaning the model's alerts are causing more problems than benefits.
- AUROC shows prediction ability, but Utility Score [5] shows whether the model is actually useful in real clinical practice.

The field has a failure it cannot see with its standard metrics.

II. Background: Limitations of the Conventional Approach

The traditional approach is:

Choose one Sepsis-3 definition → use one train/test split → report AUROC → compare models.

Four key limitations:

1. **The sepsis label can change.** Cohen et al. [6] showed that different ways of applying Sepsis-3 criteria to the same database can create very different patient groups (867–2,178 cases). The choice of label can affect results as much as the choice of model.
2. **The data split can give unrealistic results.** Guo et al. [7] found that sepsis prediction performance drops significantly when tested on data from a different time period. Random train/test splits may make models look better than they actually are.
3. **AUROC does not show the full picture.** Wang et al. [4] showed that a model can maintain a good AUROC while its clinical usefulness decreases. A high AUROC does not always mean the model helps doctors.
4. **The label may represent doctor decisions, not only patient deterioration.** Kamran et al. [8] suggested that models trained on clinical data may learn patterns of medical actions instead of learning the actual worsening of a patient's condition.

II. Background: Where Performance Variation Comes From

However, research shows that these choices can strongly affect the results:

- The choice of sepsis label can change model performance as much as changing the model itself [6].
- The way data is divided into training and testing sets can decide whether the reported performance is realistic or misleading [7].
- The choice of evaluation metric can determine whether a model failure is detected or hidden [4].

Performance differences may come more from research decisions than from the AI model itself.

This study measures how much performance variation comes from the model, the label, and the analyst's choices.

III. Research Question and Hypotheses

Research Question

In real-time sepsis prediction using MIMIC-IV [9], how much of the model's performance difference comes from: (a) the choice of AI model, and (b) the researcher's decisions, including how sepsis is defined, how data is split, when prediction is made, and how performance is measured?

Pre-Registered Hypotheses

- **H1:** Label choice affects AUROC as much as or more than model choice [6].
- **H2:** Feature importance changes across different label definitions [10].
- **H3:** Removing post-treatment information lowers AUROC by ≥ 0.03 [8].
- **H4:** Temporal testing gives lower AUROC than random testing by ≥ 0.02 [7].
- **H5:** Utility Score drops more than AUROC under realistic testing [4].
- **H6:** Statistical models perform close to gradient boosting models (within 0.02 AUROC) [11].

IV. Materials: Study Design

- **Development dataset:** MIMIC-IV v3.1 [9] (65,078 adult ICU stays)
- **External validation:** eICU-CRD v2.0 [12] (200+ US ICUs, 7.9 million person-hours)

Three ways to define sepsis [1]:

Variant	Meaning	When infection is confirmed	SOFA baseline
A	Strict	Antibiotics and blood culture must happen within 24 hours of each other.	SOFA score at ICU admission
B	Standard	Antibiotics can come up to 72 hours before the blood culture, or 24 hours after it.	SOFA score at ICU admission
C	Broad	Antibiotics and blood culture can happen within 72 hours of each other.	Lowest SOFA score in the previous 24 hours

In all three definitions, a patient has sepsis if:

There is **suspected infection**, and the **SOFA (Sequential Organ Failure Assessment)** score **increases by at least 2 points**.

V. Methods: Primary Prediction Model

The main model is a **dynamic discrete-time competing-risks hazard model**.

- It predicts the **risk of sepsis every hour** during a patient's ICU stay.
- It is built using **multinomial pooled logistic regression**.
- D'Agostino et al. [13] showed that, under short time intervals, it gives results similar to a **time-dependent Cox regression**.
- Unlike the standard Cox model, it **does not require the proportional hazards assumption**.
- It also accounts for patients who **leave the ICU or die before developing sepsis** (competing risks).

Models used for comparison

Gradient-Boosted Trees (GBT), Static Cox model, NEWS2, qSOFA, and SIRS.

V. Methods: Validation Strategy

Validation	What was done	Why?
Internal	Trained on patients from 2008–2016 and tested on 2017–2019 data [7].	To evaluate the model on newer, unseen patients.
Random split	Trained on 75% of patients and tested on 25%, chosen randomly [7].	To quantify how much the data split inflates reported performance (H4).
Pre-treatment	Used only data collected before the first antibiotic or blood culture [8].	To check whether the model is predicting sepsis or simply detecting when doctors start treatment.
External	Tested the same trained model on the eICU-CRD [12] dataset without retraining.	To see whether the model works in other hospitals.

Main evaluation metrics: **Utility Score** [5] and **Calibration**.

AUROC was reported only for comparison because it may not detect clinically important performance failures [4].

V. Methods: Methodological Commitments

- **Pre-registered protocol:** The study plan was fixed before model development to reduce bias (following TRIPOD+AI guidelines [14]).
- **Leakage prevention:** The model only uses information available at each prediction time; missing values are handled using past information only.
- **Patient-level separation:** The same patient never appears in both training and testing data.
- **Frozen external validation:** The model trained on MIMIC-IV is directly tested on eICU-CRD without retraining.
- **Reproducible pipeline:** The complete workflow uses organised scripts, shared configurations, and publicly available code for replication.

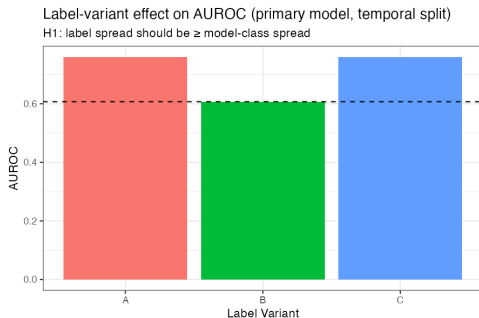
VI. Results: Cohort and Label Variance

Variant	Name	ICU Stays	Sepsis Cases	Percentage
A	Narrow	65,078	6,252	9.6%
B	Seymour-Standard	65,078	6,447	9.9%
C	Liberal	65,078	12,581	19.3%

The Liberal definition found almost **2 times more sepsis cases** than the other two definitions. The patient data and database were the same. Only the way sepsis was defined was different.

- Narrow: Strict rules → fewer patients are labeled as sepsis.
- Seymour-Standard: Common/standard Sepsis-3 interpretation.
- Liberal: More flexible rules → more patients are labeled as sepsis.

VI. Results: H1 — Label Has More Impact Than Model (CONFIRMED)



- **Label variation:** AUROC changed by **0.153** when the sepsis definition changed while using the same model.
- **Model variation:** AUROC changed by **0.145** when the model changed while using the same label.
- This confirms Cohen et al.'s [6] finding that sepsis label definition can strongly influence model performance.

The way sepsis is defined can affect performance as much as, or more than, the AI model itself.

VI. Results: H2 — Coefficient Instability (CONFIRMED)

- **10 out of 25 features** changed their effect or strength significantly across different sepsis label definitions.
- Affected features include: MAP (**Mean Arterial Pressure**), temperature, bilirubin, platelets, vasopressors, norepinephrine/epinephrine, respiratory rate change, platelet measurement, WBC measurement, and P/F ratio.

Implication

A published hazard ratio for MAP (**Mean Arterial Pressure**) could show a positive, negative, or almost no association with sepsis risk depending only on how the antibiotic-culture time window is defined [10].

Feature effects cannot be reliably interpreted without knowing which sepsis label definition was used.

VI. Results: H3 — Treatment Leakage and Label Construction

The treatment-anchored window [8] uses only information available **before the first antibiotic or blood culture**.

Metric	Value
Patient-hours analysed	179,003
Sepsis onset events	0

No sepsis onset event is found before the clinical actions used to define the Sepsis-3 label.

This is not just a small drop in model performance (0.03 AUROC). The prediction target is completely missing in the clinically realistic time window.

Most MIMIC-IV sepsis prediction models are learning when doctors suspect sepsis and start treatment, rather than detecting when a patient's health is starting to get worse.

VI. Results: H4 — Split Optimism (CONFIRMED)

Split	Model	AUROC	Utility Score
Temporal	Primary	0.607	-1.004
Random	Primary	0.730	-1.000
Temporal	GBT	0.746	-4.849
Random	GBT	0.720	-5.167

- **Primary model:** Random-split AUROC was **0.123 higher** than temporal-split AUROC (threshold: ≥ 0.02).
- The random split lets future patient patterns leak into training, inflating reported performance.
- This confirms Guo et al.'s [7] finding: **studies using random splits may overestimate real-world performance.**

The GBT model showed a smaller but reversed split effect (temporal $>$ random by 0.026), suggesting that the tree ensemble was robust to temporal shift but the linear model was not.

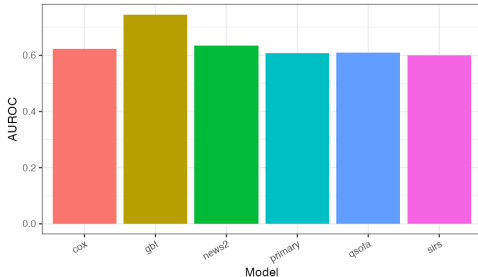
VI. Results: External Validation on eICU-CRD [12]

Variant	Model	AUROC (eICU)	AUROC (MIMIC)	Drop
A	Primary	0.736	0.760	0.024
B	Primary	0.437	0.607	0.170
C	Primary	0.725	0.759	0.034
n/a	NEWS2	0.605	0.63	0.025

- Variants A and C showed similar performance on both datasets, with only small drops.
- **Variant B (Seymour-Standard) showed a large performance drop of 0.170 AUROC when tested on eICU-CRD.**
- The most commonly used sepsis label definition did not transfer well to a different hospital dataset.

VI. Results: H6 — Model-Class Difference (**REJECTED**)

Model-class effect on AUROC (Variant B, temporal split)
H6: $\text{primary} - \text{GBT} \leq 0.02$



- **GBT AUROC: 0.746 — Primary Model AUROC: 0.607 — Difference: 0.139**
- GBT looks better based on AUROC alone.

However:

- **GBT Utility Score: -4.849 — Primary Model Utility Score: -1.004**
- The GBT model has better AUROC but **worse clinical usefulness**.

The model ranking changes depending on the evaluation metric. A model that looks better by AUROC may perform worse in real clinical use.

VI. Results: Internal Performance (Temporal Test)

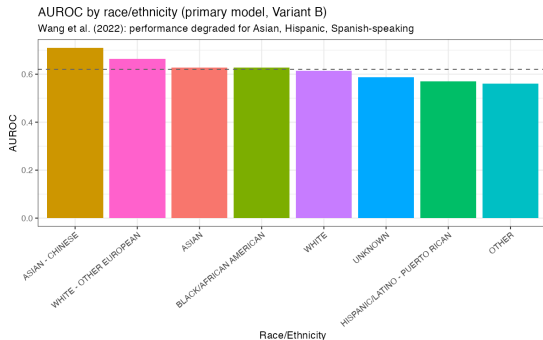
Variant	Model	AUROC	Utility Score	Alert Burden
A	Primary	0.760	−1.000	∞
A	GBT	0.742	−4.667	203
B	Primary	0.607	−1.004	∞
B	GBT	0.746	−4.849	194
C	Primary	0.759	−1.000	∞
C	GBT	0.752	−2.475	93
n/a	NEWS2	0.63–0.64	−1.1 to −1.6	97–223

All models had a negative Utility Score, meaning they did not provide clinical benefit and performed worse than giving no alerts.

Although AUROC values look reasonable, the models produced too many false alarms.

Alert burden: For every real sepsis alert, the system generated **93–359 false alerts** (benchmark: 1.4 [15]).

VI. Results: Equity Analysis (Novel Finding)

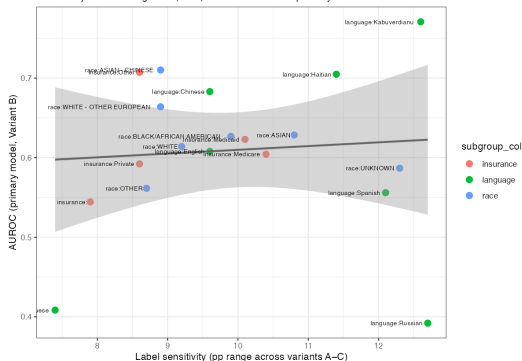


AUROC varied by up to **0.15** across racial subgroups (primary model, Variant B), from 0.56 (Other) to 0.71 (Asian-Chinese).

VI. Results: Label Sensitivity by Subgroup

Label sensitivity vs. AUROC by subgroup

Novel conjunction: Wang et al. (2022) established each separately



The percentage of patients labelled as septic changed by **7.4–14.2 percentage points** across different label definitions within demographic subgroups.

- The **Unknown/Unable-to-obtain** race groups showed the largest changes (12.3–12.8 pp).
- This is the first study to examine both model performance disparities and label sensitivity across subgroups [11].

The patients identified as septic can change depending on how the label is defined. This creates a potential fairness issue because the label itself may affect different groups differently.

VI. Results: Performance Variation by Source

Factor	AUROC Change	Related Hypothesis	Status
Sepsis label definition	0.153	H1	CONFIRMED
Model choice	0.145	H6	REJECTED
Split method (temporal vs random)	0.123	H4	CONFIRMED
Evaluation metric (AUROC vs Utility Score)	0.170	H5	n/a

All four analyst decisions create AUROC changes of 0.12–0.17.

The choice of **model** and the choice of **sepsis label definition** have a similar impact on results. Changing only one while keeping the other fixed can significantly change the evaluation.

Most studies fix one sepsis label and compare models. This measures only one source of variation while ignoring three other sources of similar importance.

VII. Discussion: Key Finding

The performance collapse reported by Wang et al. [4] is not only a model evaluation issue. This study shows that it is strongly related to how the sepsis label is constructed.

The Sepsis-3 label in MIMIC-IV has three important limitations:

1. **Depends on clinical actions** — No sepsis cases are identified before treatment starts [8].
2. **Sensitive to label definition** — Different reasonable interpretations can produce almost double the number of sepsis cases [6].
3. **Produces poor clinical utility** — All tested models show negative Utility Scores [4].

These findings suggest that improving model complexity alone cannot solve the problem.
Better sepsis label design is required.

VII. Discussion: Contributions

#	Contribution
C1	Compared three different Sepsis-3 label definitions using the same prediction model on MIMIC-IV [6].
C2	Showed that no sepsis cases exist before antibiotics or culture tests, highlighting that the label depends on clinical actions [8].
C3	Confirmed that all models had negative Utility Scores, extending the findings of Wang et al. [4].
C4	Found that 10 clinical features changed their effects across different label definitions, supporting Lauritsen et al. [10].
C5	Quantified split optimism: random splits inflated AUROC by 0.123 compared to temporal splits, confirming Guo et al. [7].
C6	First study to examine how different sepsis label definitions affect racial subgroup analysis [11].
C7	Used a pre-registered, TRIPOD+AI-compliant, fully reproducible research pipeline [14].

VII. Discussion: Limitations

- **Small external event count:** The eICU dataset had only **63 sepsis cases**, making some evaluation metrics less reliable.
- **Deep learning comparison:** A deep survival model was planned but not implemented.
- **Limited features:** Clinical notes, procedure codes, and ventilator settings were not included.
- **Specific to Sepsis-3:** These findings apply to the Sepsis-3 label and may not apply to other sepsis definitions.

These limits are documented rather than hidden.

VIII. Conclusion and Future Work

1. **Better sepsis labels:** Define sepsis using objective patient measurements (such as lactate levels and organ function) instead of treatment decisions.
2. **Prospective study:** Test the new label in real clinical settings using future patient data.
3. **Better alert threshold:** Choose the alert threshold that maximises clinical benefit (Utility Score) instead of only improving prediction accuracy.
4. **Multi-hospital evaluation:** Test the model in different hospitals and identify an acceptable number of false alerts for clinical use.

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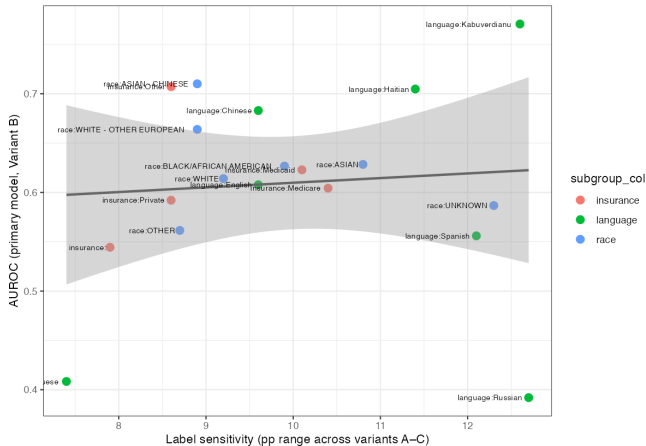
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Appendix: Label Sensitivity vs. AUROC

Label sensitivity vs. AUROC by subgroup

Novel conjunction: Wang et al. (2022) established each separately



Thank You

Questions & Discussion

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M.Sc. in Computing · Atlantic Technological University (ATU), Galway

Pre-registered protocol. All extraction, modelling, and evaluation code released for reproducibility.